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SOMATOSENSORY PATHWAYS

----- Active space --

Levels of control for general sensations :

Skin

↓

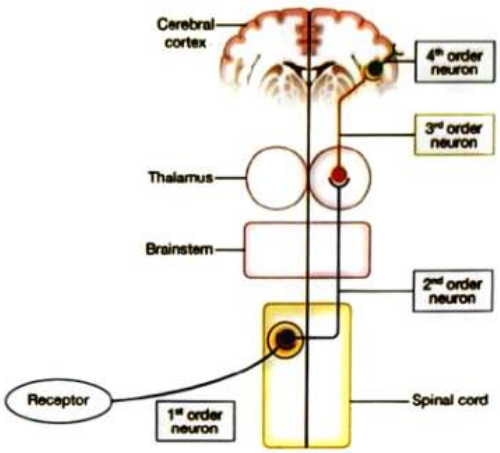
Spinal cord

↓

Thalamus

↓

Sensory cortex.



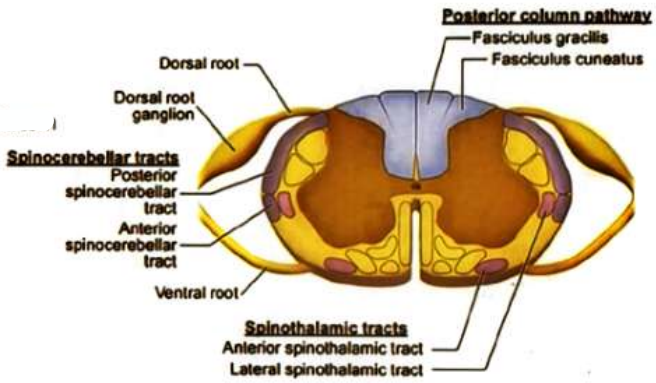
Spinothalamic Pathways

00:01:05

From spinal cord to thalamus.

PATHWAYS :

1. Posterior column pathway (Dorsal column-[illegible] pathway).
2. Anterolateral spinothalamic pathway.
3. Spinocerebellar tract.

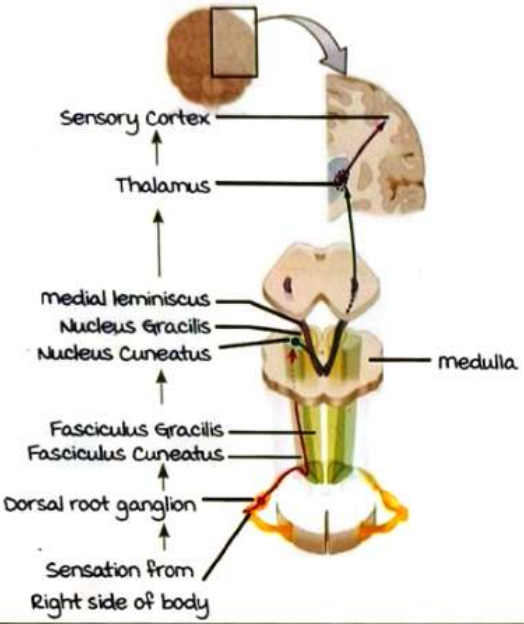
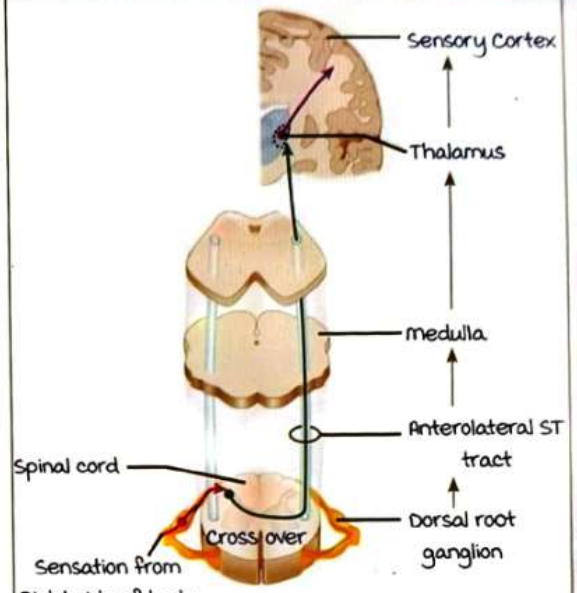
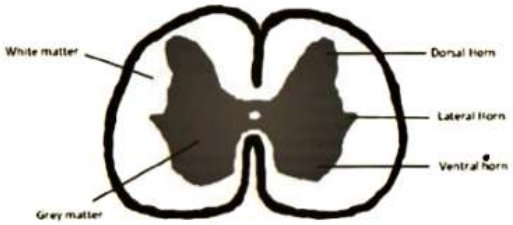
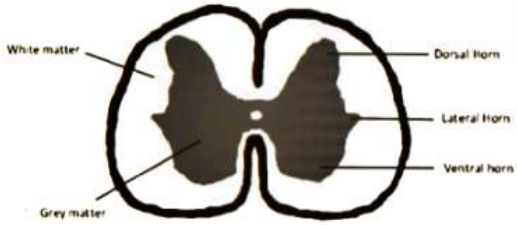


Somatosensory pathway

	Dorsal column medial lemniscal pathway	Anterolateral pathway.
Myelination	Myelinated (Majority)	Unmyelinated
According to Evolution	New	Old
Conduction Speed	Fast	Slow
Sensations	• Conscious Proprioception (Joint position sense). • Fine touch. • Vibration. • Stereognosis (Ability to identify objects by touch on eyes closed).	• Lateral Pathway : Pain. • Anterior Pathway : – Crude touch. – Pressure. • Temperature. • Itch.
Level of Decussation	Medulla	Spinal cord

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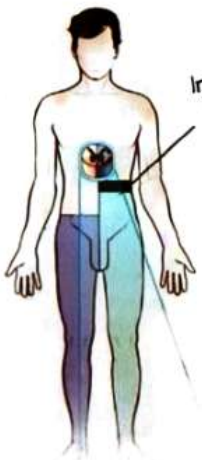
----- Active space -----

	Dorsal column medial lemniscal pathway	Anterolateral pathway
Pathway		
	Lower limb : Gracile Fasciculus → Nucleus Gracilis Upper limb : Cuneate Fasciculus → Nucleus Cuneatus	
Somatotopic arrangement C : Cervical T : Thoracic L : Lumbar S : Sacral	(medial) SLTC (Lateral) 	(medial) CTLS (Lateral) 

APPLIED ASPECT :

Brown - Sequard Syndrome :

Hemisection of spinal cord.



Brown Sequard Syndrome

Feedback

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----- Active space -----

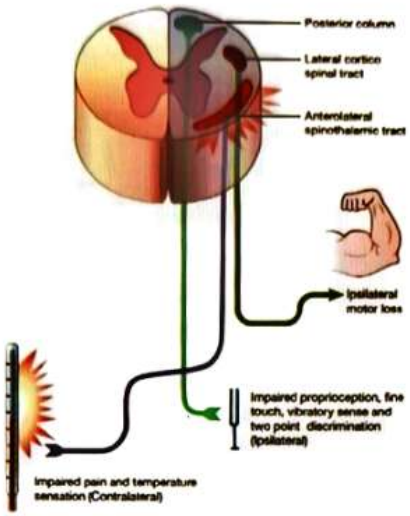
Clinical manifestations :

Ipsilateral :

- Loss of Dorsal column sensations :
  - Proprioception.
  - Vibration.
  - Fine touch.
- motor paralysis.

Contralateral :

- Loss of Anterolateral column sensations :
  - Pain.
  - Temperature.



Thalamus

00:19:37

Hallmark : Sensory relay station.

Order of neurons : 3rd order.

Nuclei : Ventropostero-lateral (VPL), for general senses : Touch, pain, temperature.

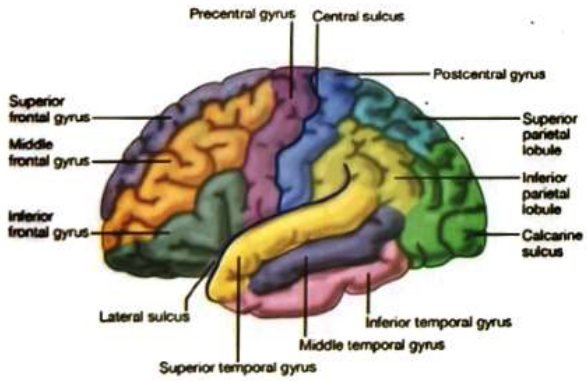
Clinical significance :

Dejerine-Roussy syndrome (Thalamic pain syndrome) :

- Pathology : Stroke (Infarction) → Nerve fibres in VPL nuclei are inrritable → ↑Neuronal signalling.
- Symptom : Excruciating pain.

Sensory Cortex

00:23:06



Cerebral Cortex

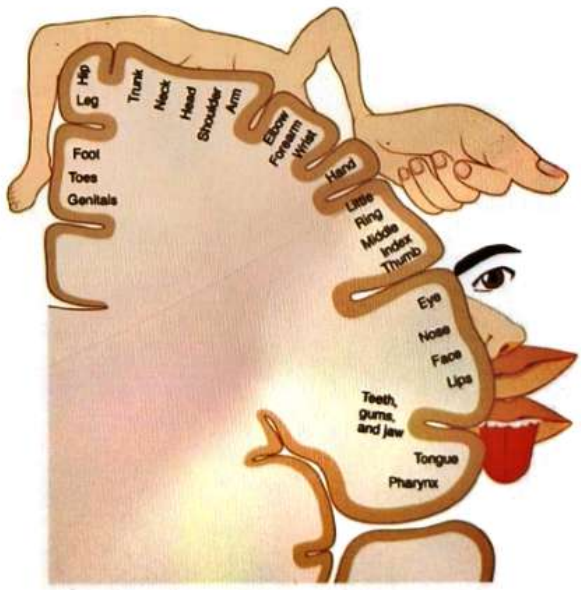
Location :

Postcentral gyrus (Parietal cortex) : Broadmann areas 3, 1, 2.

Note : motor cortex is precentral gyrus.

Function :

Localization of sensation



Penfield sensory homunculus

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----- Active space -----

SENSORY HOMUNCULUS :

Representation :

- Medial : Leg.
- Lateral : Face.
- Maximum: Thumb, lip (Based on use).
- Minimum : Hip, back, trunk.
- No representation of visceral organs.

Note :

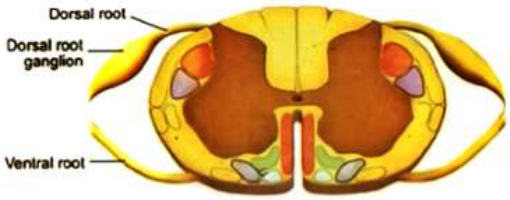
Referred pain : Visceral organ pain is referred to somatic structure with same embryonic origin.

Laws In Sensory Physiology 00:32:13

BELL-MAGENDIE LAW :

Spinal dorsal root : Sensory.

Spinal ventral root : Motor.



Spinal cord

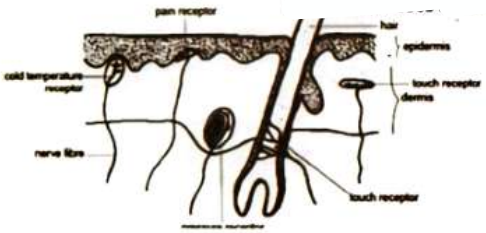
LAW OF SPECIFIC NERVE ENERGIES :

AKA Muller's Doctrine.

Sensory receptors detect specific sensations.

Eg :

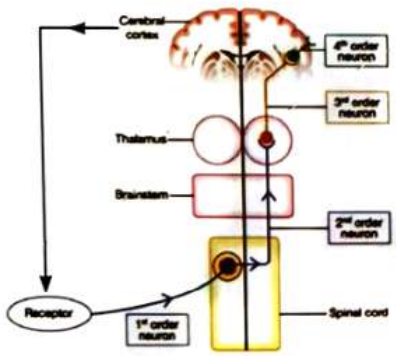
- Pain receptors : Pain.
- Touch receptors : Touch.
- Cold temperature receptors : Cold.



Sensory Receptors

LAW OF PROJECTION :

On receptor stimulation, response of cerebral cortex is projected to lowest receptor level.



Projection pathway

Cortical Plasticity :

Reorganization of cortical representation by adjacent areas following injury.

Clinical significance : Phantom limb pain.

Eg : Hand amputation → Cortical takeover by arm → Arm pain → Perceived as hand pain.

WEBER-FECHNER LAW :

Magnitude of sensation felt is directly proportional to log intensity of initial stimulus.

- Eg :



Feedback